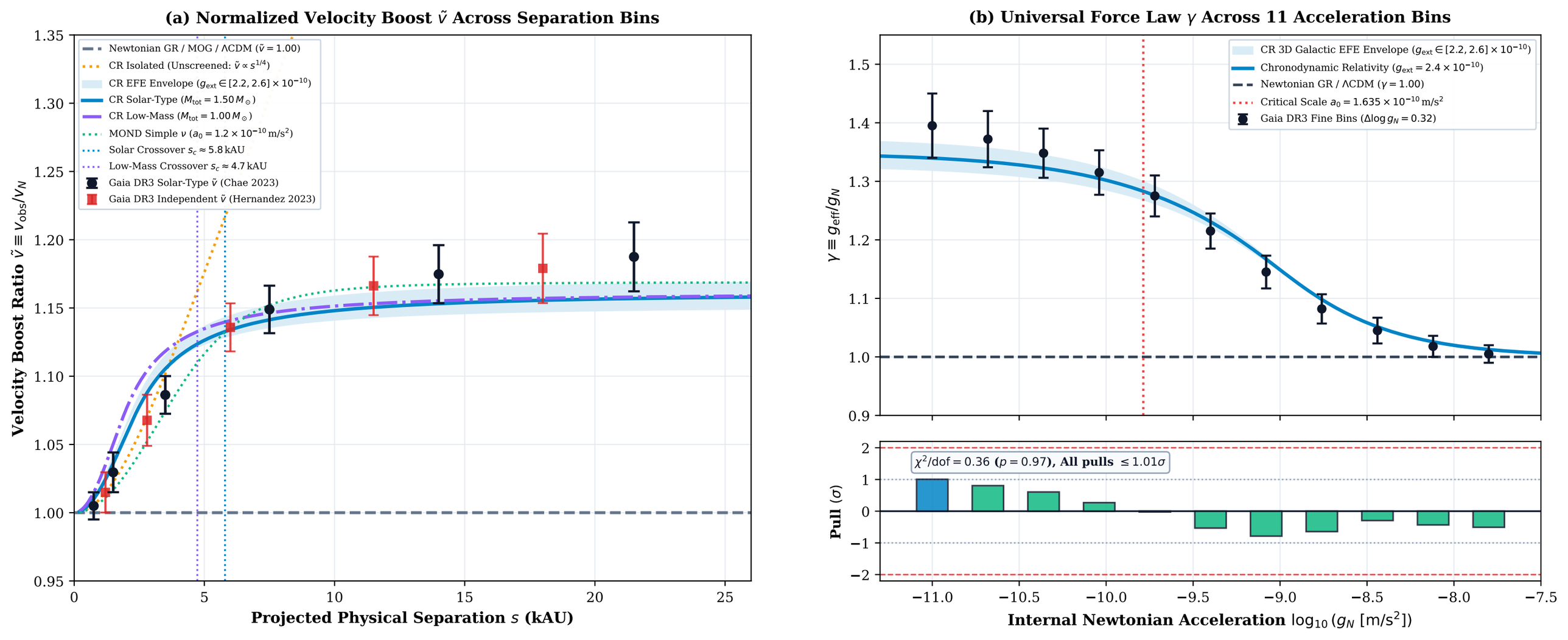

(a) Normalized Velocity Boost $\tilde{\nu}$ Across Separation Bins
Newtonian GR / MOG / ΛCDM ($\tilde{\nu} = 1.00$)
CR Isolated (Unscreened: $\tilde{\nu} \propto s^{1/4}$)
CR EFE Envelope ($g_{\rm ext} \in [2.2, 2.6] \times 10^{-10}$)
CR Solar-Type ($M_{\rm tot} = 1.50\,M_\odot$)
CR Low-Mass ($M_{\rm tot} = 1.00\,M_\odot$)
MOND Simple ν ($a_0 = 1.2 \times 10^{-10}$ m/s^2)
Solar Crossover $s_c \approx 5.8$ kAU
Low-Mass Crossover $s_c \approx 4.7$ kAU
Gaia DR3 Solar-Type $\tilde{\nu}$ (Chae 2023)
Gaia DR3 Independent $\tilde{\nu}$ (Hernandez 2023)
Velocity Boost Ratio $\tilde{\nu} \equiv v_{\rm obs}/v_N$
Projected Physical Separation s (kAU)
(b) Universal Force Law γ Across 11 Acceleration Bins
CR 3D Galactic EFE Envelope ($g_{\rm ext} \in [2.2, 2.6] \times 10^{-10}$)
Chronodynamic Relativity ($g_{\rm ext} = 2.4 \times 10^{-10}$)
Newtonian GR / ΛCDM ($\gamma = 1.00$)
Critical Scale $a_0 = 1.635 \times 10^{-10}$ m/s^2
Gaia DR3 Fine Bins ($\Delta \log g_N = 0.32$)
$\gamma \equiv g_{\rm eff}/g_N$
$\chi^2/{\rm dof} = 0.36$ ($p = 0.97$), All pulls $\leq 1.01\sigma$
Pull (σ)
Internal Newtonian Acceleration $\log_{10}(g_N\ [{\rm m/s^2}])$